FINAL VIVA · ANLP MONSOON 2026

C3: Committed Confidence Calibration

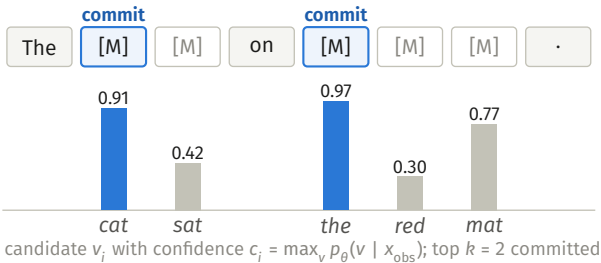
Selection bias and two sources of error in confidence-based token commitment in diffusion language models

Abhiraj Ratna · Ayush SM · Ojas Kataria · Parth Sharma

Team DuckDuckDuckGoose

Diffusion LMs commit the tokens they are most confident about

One forward pass scores every masked position; the decoder keeps the top-k



- LLaDA-8B and Dream-7B keep the most confident positions (MaskGIT-style); Fast-dLLM puts one global threshold on the same number.
- A committed token won: its confidence was the largest of m noisy estimates. That is the setting of the winner's curse (Efron 2011; Andrews et al. 2024).

Question. Once a token has won, is its confidence still an honest probability of being right? And when several tokens win in one step, which error dominates?

Two separate ways parallel decoding goes wrong

For a set S committed in one step, given the observed context x_{obs}

$$\mathrm{KL}\Big(p^* \,\Big\|\, \prod_{i \in S} p_{\theta,i}\Big) = \underbrace{\mathrm{TC}(x_S \mid x_{\text{obs}})}_{\text{(A) dependency}} + \underbrace{\sum_{i \in S} \mathrm{KL}\big(p_i^* \,\|\, p_{\theta,i}\big)}_{\text{(B) estimation}}$$

- **(A)** Marginals, never the joint: tokens can each be plausible yet jointly nonsensical.
- **(B)** The marginals themselves are wrong. The dependency literature treats **(B)** as fixed.
- C2 measures both where p^* is known; C1 audits calibration under selection.

Work	Calibrated under selection?	Separates (A) and (B)?
Fast-dLLM; OSDT	no	no
KLASS; SWD; LESS	no	no
PC-Sampler	positional only	no
DAPD; DEMASK; PUNT	no	(A) only
ParallelBench	no	(A) only
Tang et al. (2026)	no	identifies
This work	**yes**	**both**

Three contributions

C1

Selection-conditional calibration audit

Is a token that *won* its pool overconfident? On LLaDA-8B, Dream-7B (Base and Instruct) and a 110M masked diffusion LM we trained with our evaluation documents held out. Pre-registered; numbers from a held-out report half.

C2

Both terms of the KL split

Exactly, in nats, on a synthetic JSON probe where p^* is known. On real text, where it is not, proxies whose behaviour we first check on the probe: a direction, not a split in nats.

C3

A recalibrated commit rule

A threshold on recalibrated confidence, built only from what the decoder already computes (no extra forward passes), against six decoders at matched tokens per step on GSM8K and MBPP.

Order of the talk. Theory and the instrument, then C1, C2 and C3 in turn, then what we got wrong and what remains open.

Theory: where a winner's curse can and cannot come from

State s: a document with masked set M. Teacher forcing makes correctness $a_i = 1\{v_i = x_{0,i}\}$ observable

Proposition 1 · Winning adds nothing given confidence and context

If the pairs (ℓ_j, a_j) are independent across $j \in M$ given s, and the pool P is drawn independently of them, the winner i of P satisfies

$$\Pr\left(a_i = 1 \mid \ell_i,\ s,\ i = \arg\max_{j \in P} \ell_j\right) = \pi(\ell_i, s).$$

However unequal the noise scales: heteroskedastic noise alone cannot bias committed confidence. This corrected the proposal's argument.

Proposition 2 · Exact split of the committed-token gap

$$\Delta(m) = \underbrace{\mathbb{E}_m[\sigma(\ell) - \pi(\ell)]}_{\text{own miscalibration}} + \underbrace{\mathbb{E}_m[\pi(\ell) - \pi(\ell, s)]}_{\text{field term}}$$

$\pi(\ell)$: ordinary calibration curve; $\pi(\ell, s)$: its context-aware version. With a pool of one, winners are random, so the field term is exactly 0.

The field term is nonzero only when accuracy at fixed confidence depends on context: the **winner's curse in a corrected, testable form**. Proofs in the backup.

C1: what we measure, and when we declare an effect

Sign: confidence − accuracy, so positive means overconfident (the proposal used the opposite)

E1 · committed-token gap

$\Delta(m) = \mathbb{E}_m[c_i - a_i]$ over tokens that win pools of size $m \in \{1, 2, 4, 8, 16, 32\}$. Primary contrast $\Delta(32) - \Delta(1)$.

E2 · field effect

Observed $\Delta(m)$ minus the gap predicted by $\hat{\pi}(\ell)$, which sees only the token's own confidence. A context-aware $\hat{\pi}(\ell, s)$ should cover the rest.

E3 · trajectory shift

Training uses uniform masks; decoders fill easy positions first. Are decoder states less calibrated at equal confidence and matched mask rate?

Hypotheses. **H1** $\Delta(32) - \Delta(1) > 0$; **H2** the field term at $m = 32$ is positive; **H3** decoder states are more overconfident than uniform ones.

Declare rule. The 95% interval excludes zero *and* the estimate exceeds twice the non-determinism floor. The floor is 0 for all three model families, so $\delta_{\min} = 0.02$, fixed in advance, sets power. Holm across models × datasets.

Diagnostics fitted on winning tokens:

$$\Pr(a_i{=}1) = \sigma\big(\alpha\,[{+}\alpha_s] + f(\ell_i) + \beta_z z_{r,m}\big),$$

or rank dummies γ_k in place of $z_{r,m}$; f a 9-knot spline fixed in advance. State effects α_s absorb the field term: $\beta_z < 0$ without them and ≈ 0 with them signals a field effect.

C1: the trace sweep

Collect once on GPU, one row per (masked state, position); draw pools and fit on CPU

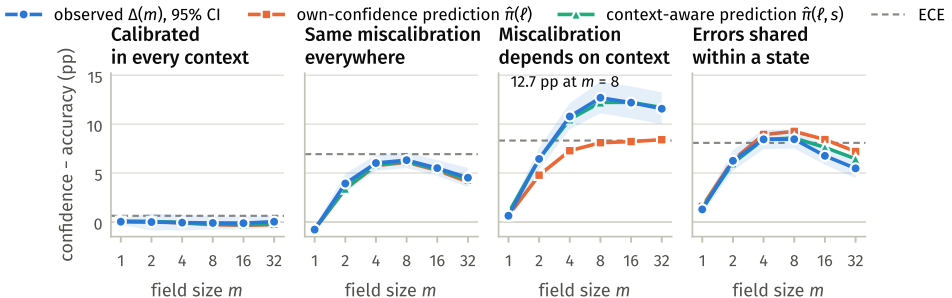
Model	Text	Prec.	Docs	Rows (M)	GPU-h
LLaDA-8B Base	FineWeb-Edu	fp16	1,979	119.6	68
Dream-7B Base	FineWeb-Edu	fp16	1,976	119.4	60
Our 110M MDLM	FineWeb-Edu	fp32	1,979	119.6	3
LLaDA-8B Instruct	GSM8K	fp16	1,319	13.8	18
LLaDA-8B Instruct	MBPP	fp16	499	3.1	5
Dream-7B Instruct	GSM8K	fp16	1,319	13.4	14
Dream-7B Instruct	MBPP	fp16	499	2.5	4
LLaDA-8B Base	time-split	fp16	500	30.2	17
Dream-7B Base	time-split	fp16	500	30.2	15
Our 110M MDLM	time-split	fp32	500	30.2	1
LLaDA-8B Base	FineWeb-Edu	4-bit	198	12.0	5

Top: the pre-registered arms. Grey: the unseen time-split set and the 4-bit control (T8). RTX 2080 Ti: 208 GPU-hours, 0.49 billion rows.

- **States per document**: uniform masks at rates 0.1–0.9; teacher-forced confidence top-k trajectories, $k \in \{2, 4, 8\}$; uniform masks matched to them (466 passes per document).
- **Per row**: candidate, confidence, logit, correctness, p(true), entropy, margin, top-20.
- **Halves**: 1,000 fit / 1,000 report documents. Calibrators and splines from the fit half; every number from the report half.
- **Deterministic**: repeats are bitwise identical, so the floor is 0 for every family.
- **Frozen first**: analysis commands tagged `prereg-v1` before any report-half number; changes go to `prereg/deviations.md`.

The instrument recovers known truth (T0)

Four simulated worlds · fit half → report half · document bootstrap



Why ECE is not enough. When miscalibration depends on context, ECE is 8.3 pp but Δ(32) − Δ(1) = 10.9 pp. Known weakness: the field-term estimator reads high (3.3 pp against a true 1.6 pp).

C1 results: does winning make a token overconfident? (E1, E2)

RESULTS LAND TUE 27 OCT

Report-half analysis: `C3_REPORT_HALF_OK=1 bash prereg/analysis_commands.sh` at tag `prereg-v1` (8B arms: Abhiraj; 110M arms: Ojas). Macros: `results/c1/<Label>/macros.tex`, prefix `COne<Label>`.

- Figure: $\Delta(m)$ against field size m, one panel per model (LLaDA-8B, Dream-7B, 110M), primary arms `uniform` and `decoder:topk_conf:k4`.
- Table: $\Delta(32) - \Delta(1)$ with 95% interval and the Holm verdict per arm (models × datasets); ECE beside it.
- E2: field term at $m = 32$ from the own and the context-aware calibrator.
- One-line answer to H1 and H2, with the null stated plainly if that is what we get.

C1 results: decoder states, tasks and controls (E3, T8)

RESULTS LAND TUE 27 OCT

Same run: `scripts/analyze_e3.py` (prefix `EThree<Label>`) and `scripts/run_controls.py` (prefix `TEight<Label>`).

- E3: decoder-induced states against matched-uniform states at equal confidence, per model; answer to H3.
- Instruct models on GSM8K and MBPP: is the committed-token gap larger on tasks than on web text?
- T8 ablation table: position, token type, per-step temperature; LLaDA-8B-Base 4-bit against fp16 on the precision subset.
- Time-split arms: the same contrasts on text the 8B models cannot have seen.

C2: a JSON probe where the true joint p^* is known

Our 110M model, fine-tuned on the probe; both terms computed exactly at every decoding step

- **Instances**: the simple fields of 200 SchemaStore schemas, 6–16 per schema, single-token values with seeded Dirichlet marginals.
- **Dependency knob** ρ: fields are paired. With probability ρ, stated in a marker at the start, the follower is a fixed function of its leader; otherwise an independent draw. So p^* is exact.
- **Model**: our 110M fine-tuned on 200k instances; evaluated on 1,000 (200 per ρ).
- **Decoding**: all values masked; each step a rule picks k fields, the most confident or at random, and we fill them with the true values.
- **Check**: **(A)** + **(B)** equals the directly computed KL; largest error 4.3×10^{-14} nats (bar 10^{-6}).

one evaluation instance (ρ = 1), values masked

```
{"_rho":"1.00",
  "bundle":[M], "path":[M],
  "ref":[M], "repo":[M],
  "stats":[M], "version":[M]}
```

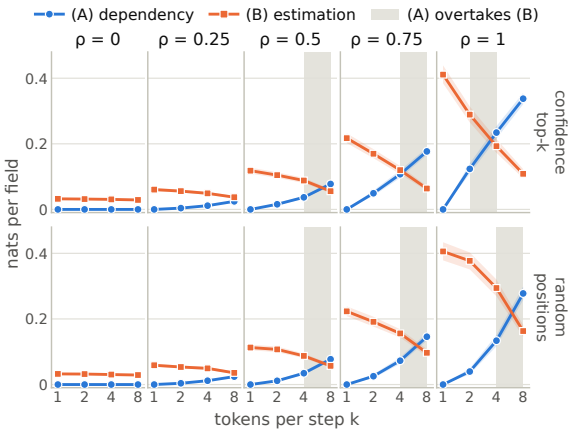
Per step, for the committed set S

(A) = $\sum_{i \in S} H(p_i^*) - H(p_S^*)$: zero unless a leader and its follower are committed together.

(B) = $\sum_{i \in S} \mathrm{KL}(p_i^* \,\|\, p_{\theta,i})$: mass off the allowed values counts against **(B)**.

Summed over steps, per field, bootstrapped over instances.

C2: dependency overtakes estimation only at high ρ and k



- **(A)** is zero at $k = 1$ and at $\rho = 0$, and grows with both.
- For $\rho \geq 0.5$, **(A)** overtakes **(B)** at $k = 8$; at $\rho = 1$ under confidence top-k already at $k = 4$. Each crossover is also where the 95% interval of **(A)** –**(B)** clears zero.
- For $\rho \leq 0.25$, **(B)** dominates up to $k = 8$.

$\rho = 1$, $k = 8$, confidence top-k

(A) 0.338 [0.320, 0.355] and **(B)** 0.109 [0.095, 0.123] nats per field: **(A)** is 76% of the total.

Main model: the fine-tune that never saw the ρ marker masked (choice in the backup). 10,000 bootstrap resamples over instances.

C2: three findings that correct our plan

1 · Confidence selection raises (A), not (B)

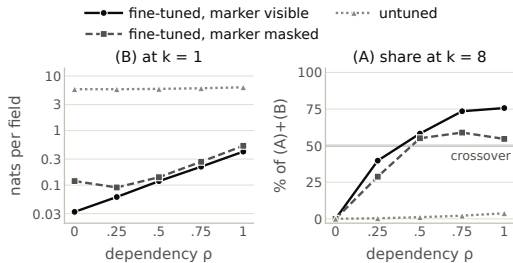
Planned: ranking on noisy confidences inflates **(B)**.
Where **(B)** is higher than at random, the intervals overlap.
(A) rises instead ($\rho = 1$, $k = 2$): 0.124 [0.108, 0.140] against 0.040 [0.031, 0.049] at random.

2 · (B) depends on ρ

Planned: ρ moves only **(A)**.
At $k = 1$, **(B)** rises from 0.032 at $\rho = 0$ to 0.411 at $\rho = 1$: the model reads the marker only in part.

3 · (A) and (B) trade off along k

At $\rho = 1$, from $k = 1$ to 8, **(B)** falls from 0.411 to 0.109 while **(A)** rises from 0 to 0.338: a follower decoded with its leader shows its error as **(A)**, not **(B)**.



- Without fine-tuning, **(B)** dominates everywhere: **(A)** is at most 4% of the total.
- The ranking commits leaders together with their followers more often than chance, which brings the crossover forward.

C2 on real text: the dependency proxy is near zero

300 fit-half FineWeb-Edu documents, one state per mask rate; the proxy is first checked on the probe

Proxies

For **(A)**: **parallel–sequential disagreement**, the fraction of the k most confident positions whose token changes when committed one at a time rather than in one pass. For **(B)**: the gap $\Delta = c - a$ and the cross-entropy (CE) of the true token.

On the probe, it tracks (A) weakly

Spearman with the exact **(A)** over the 15 confidence cells with $k \geq 2$: 0.49 [0.28, 0.63]; 0.28 for the other fine-tune. It rises with k, barely with ρ: at $k = 8$, 0.028 at $\rho = 0$, where **(A)** is zero, and 0.049 at $\rho = 1$.

mask rates 0.1–0.9	110M (fp32)	LLaDA-8B Base (fp16)
disagreement, $k = 4$	0.0003	0.0001
disagreement, $k = 8$	0.0021	0.0022
CE (nats), $k = 1 \rightarrow 8$	0.08 → 0.40	0.08 → 0.23
Δ, $k = 1$	+0.001 [−0.003, 0.005]	+0.003 [−0.001, 0.006]
Δ, $k = 8$	+0.007 [0.004, 0.010]	+0.004 [0.002, 0.007]

CE includes the text's own entropy, and lower ranks are harder positions.

At 2–8 tokens per step, real text looks like the probe's **low-ρ regime, where (B) dominates**. But the proxy sees only dependence the model has learned: near-zero disagreement does not show that **(A)** is small.

C3: a recalibrated commit rule against six decoders

RESULTS LAND THU 29 OCT

T11 (the C3a policy and its calibrator, `artifacts/c3/`) and the T12 sweep (Parth). Macros from the T12 scoring outputs.

- C3a: commit every masked position in the active block whose recalibrated confidence clears $1 - \alpha$, and at least one. Calibrator fitted on the fit half of the LLaDA-8B-Instruct GSM8K and MBPP traces, then frozen.
- Matched-speed table: accuracy against tokens per step for top-k, the Fast-dLLM threshold, entropy-bounded, margin, KLASS, PC-Sampler and C3a, on the GSM8K and MBPP report halves; wall-clock beside it.
- Sanity row: top-k and the threshold decoder against their published numbers.
- The answer in one line, including “no gain at matched speed” if that is the result.

Summary

1 · Theory

Selection alone cannot bias a calibrated posterior (Proposition 1). Any bias lives in the field term (Proposition 2), which the instrument measures and which T0 shows it can see where ECE cannot.

2 · C1 audit

Placeholder: the answers to H1–H3 on three model families, from the report half (Tue 27).

3 · C2: (A) and (B) separated

On the probe, **(A)** overtakes **(B)** only for $\rho \geq 0.5$ and larger k; confidence selection raises **(A)**, not **(B)**. On real text at 2–8 tokens per step, the **(A)** proxy is near zero: the **(B)**-dominated regime.

4 · C3 decoding

Placeholder: C3a against six decoders at matched tokens per step (Thu 29).

What we got wrong, and the limits of these results

Claims we withdrew

- Heteroskedastic noise alone makes winners overconfident (proposal): false, Proposition 1.
- Selection on confidence inflates **(B)** (proposal, mid report): on the probe it raises **(A)** instead.
- ρ moves only **(A)** (plan v2): **(B)** rises with ρ too.
- C2 on an 8B model (plan v2): not run. The probe results are for our 110M only.
- Real-text C2 as identified contrasts (mid deck): they are proxies, with a direction but no split in nats.

Limits

- C2's exact values hold for one synthetic distribution and one 110M model.
- The main probe model was chosen by a fallback (lower validation NELBO) after neither fine-tune met a bar that was itself written down with the results: **not pre-registered**. The other model's crossovers are in the backup.
- The disagreement proxy tracks **(A)** weakly, and mostly through k.
- 8B models in fp16, not bfloat16 (no bfloat16 on our GPUs); E3 for top-k decoders only (`prereg/deviations.md`).
- C1 and C3 limits: pass 2.

Thank you. Questions?

Code	github.com/abhirajratna/ANLP-Project
Weights & Biases	wandb.ai/ANLP-project/c3-dllm-calibration
Hugging Face	huggingface.co/ANLP-project
Paper	DuckDuckDuckGoose-Final.zip

Backup: proofs

W_i: the event that $i \in P$ wins, $i = \arg\max_{j \in P} \ell_j$; ties broken by independent randomness U

Proposition 1

Suppose, given s, each pair (ℓ_i, a_i) is independent of $\ell_{-i} = (\ell_j)_{j \in M \setminus \{i\}}$. Let $Z = (\ell_i, s)$ and $Y = (\ell_{-i}, P, U)$.

Given s, $(\ell_i, a_i) \perp \ell_{-i}$ and $(P, U) \perp (\ell_j, a_j)_{j \in M}$, so $(\ell_i, a_i) \perp Y \mid s$, and by weak union $a_i \perp Y \mid Z$, i.e. $\mathbb{E}[a_i \mid Y, Z] = \pi(\ell_i, s)$.

W_i is determined by (Y, Z), so for every bounded measurable h

$$\mathbb{E}[a_i h(Z) 1_{W_i}] = \mathbb{E}[\pi(\ell_i, s) h(Z) 1_{W_i}].$$

Dividing by $\Pr(W_i) > 0$ gives $\mathbb{E}[a_i \mid Z, W_i] = \pi(\ell_i, s)$ a.s.

Proposition 2

Exactly one W_i holds. Applying the display with $h \equiv 1$, summing over i and averaging over s:

$$\mathbb{E}_m[a] = \sum_i \mathbb{E}_m[a_i 1_{W_i}] = \sum_i \mathbb{E}_m[\pi(\ell_i, s) 1_{W_i}] = \mathbb{E}_m[\pi(\ell, s)],$$

so $\Delta(m) = \mathbb{E}_m[\sigma(\ell) - \pi(\ell, s)]$; add and subtract $\pi(\ell)$.

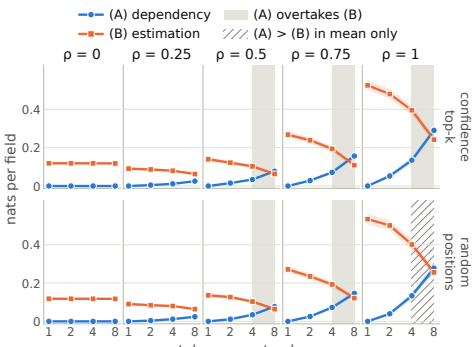
At $m = 1$, $\mathbb{E}_1[\pi(\ell)] = \mathbb{E}_1[a]$ by the tower property and $\mathbb{E}_1[\pi(\ell, s)] = \mathbb{E}_1[a]$ by the display, so the field term is 0.

Backup: how the main probe model was chosen

Recorded in `artifacts/probe/main_variant.json`

	ft110m-v3-visible	ft110m-v3
ρ marker in fine-tuning	never masked	masked
Fine-tuning tokens (M)	300	100
Marker effect on copy	0.251	0.014
Val. NELBO (nats/tok.)	0.1751	0.1892
First k with **(A)** > **(B)** (ρ=1)	4	8

- The bar for the main result, a marker effect of 0.5, was first written down *with* these results. Neither model meets it.
- The fallback, the lower validation NELBO, is stated as a rule in no committed file; it was applied when the choice was made. **Not pre-registered.**
- The other model crosses over at $k = 8$ for every $\rho \geq 0.5$ under both rules: the qualitative picture holds.



As the main figure, for ft110m-v3. Hatched: **(A)** > **(B)** in mean only.

Backup: disagreement by mask rate (T10)

Disagreement at $k = 8$ per mask rate; one state per document per rate

Mask rate	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
110M	0.0004	0	0	0	0	0	0	0.0013	0.0171	0.6250
LLaDA-8B-Base	0	0	0	0	0	0	0	0.0008	0.0192	0

- At rate 1 every input is the same all-`[MASK]` sequence. That one state gives the 110M model nearly all of its disagreement (0.0644 at $k = 8$ over all ten rates); LLaDA does not disagree there. The main slide reads rates 0.1–0.9.
- Documents under 256 tokens are skipped (a deviation from the plan); at least 8 masked positions per state; 10,000 document-bootstrap resamples.

Backup: data, contamination and our 110M model

- **2,000 calibration documents**, reservoir-sampled from FineWeb-Edu with a fixed seed and excluded from our 110M's training stream; 1,000 fit / 1,000 report.
- **Tasks**: GSM8K and MBPP test sets, split 50/50 the same way, as teacher-forced documents.
- **Contamination screen (T4)**: prompt with 50 tokens, continue greedily, flag a verbatim run ≥ 32 tokens. Flagged: 2/1,979 (LLaDA), 1/1,976 (Dream); the 8B arms are also analysed without them.
- **Time-split set**: text written after the 8B models were trained.

Model	bidirectional transformer: 12 layers, width 768, 12 heads; sequence length 256
Objective	MDLM (weighted masked cross-entropy), GPT-2 BPE + [MASK]
Training	169,055 steps, 10.0B tokens of FineWeb-Edu

37.2

perplexity *bound* on the 2,000 held-out docs [36.5, 38.0]

28.5

GPT-2 small, *exact* perplexity, same docs [28.0, 29.0]

MDLM gives only an upper bound, and GPT-2 saw other data with far more compute.

Backup: references

Andrews, Kitagawa & McCloskey (2024). Inference on winners. *Quarterly Journal of Economics* 139(1).
Azangulov et al. (2026). Parallel sampling from masked diffusion models via conditional independence. *ICLR*.
Ben-Hamu et al. (2025). Accelerated sampling from masked diffusion models via entropy bounded unmasking. *NeurIPS*.
Chang et al. (2022). MaskGIT: masked generative image transformer. *CVPR*.
Cobbe et al. (2021). Training verifiers to solve math word problems. arXiv:2110.14168.
Efron (2011). Tweedie's formula and selection bias. *JASA* 106(496).
Guo et al. (2017). On calibration of modern neural networks. *ICML*.
Kang et al. (2026). ParallelBench: understanding the trade-offs of parallel decoding in diffusion LLMs. *ICLR*.
Kim et al. (2026). DAPD: dependency-aware parallel decoding via attention for diffusion LLMs. *ICML*.
Nie et al. (2025). Large language diffusion models. *NeurIPS*.
Penedo et al. (2024). The FineWeb datasets: decanting the web for the finest text data at scale. arXiv:2406.17557.
Ringel, Ali & Romano (2026). Dependency-guided parallel decoding in discrete diffusion language models. arXiv:2604.02560.
Sahoo et al. (2024). Simple and effective masked diffusion language models. *NeurIPS*.
SchemaStore contributors (2026). SchemaStore. github.com/SchemaStore/schemastore.
Tang et al. (2026). Is your diffusion sampler actually correct? *ICML*.
Wu et al. (2025). Fast-dLLM: training-free acceleration of diffusion LLM. arXiv:2505.22618.